

Aditya Khamitkar

+91-93907-99252 · khamitkaraditya@gmail.com · linkedin.com/in/adityakhamitkar · github.com/TechieSamosa · kaggle.com/flickeringtubelight

SUMMARY

Data Scientist and AI/ML Engineer with expertise in production LLM pipelines, advanced statistical analysis, and predictive modelling. Demonstrated success driving data-quality audits, distributed ETL, and end-to-end ML research from exploratory analysis to deployed models to deliver actionable insights. Experience spans 500+ hours of RLHF evaluation, agentic AI systems, time-series forecasting, and geospatial deep learning.

EDUCATION

Vellore Institute of Technology (VIT) Sep 2022 – Jun 2027
Integrated M.Tech., Computer Science & Engineering ; Specialisation: Computational & Data Science

EXPERIENCE

AI Training Specialist & Data Science Consultant Jan 2025 – Present
Deccan AI Experts (Projects: MAITRIX, Shield, Bluebird, Action Extensions) *Remote*
– Boosted Google Gemini instruction-following accuracy by **15%** by engineering structured RLHF evaluation rubrics and grading frameworks for **4 frontier LLMs**.
– Eliminated RLHF data integrity violations across **3 production corpora** via systematic EDA, distribution analysis, and outlier detection using Pandas and NumPy.
– Resolved **100%** of schema bugs in an **18-tool** agentic RL environment (Project MAITRIX) by automating Python-based SQLite database audits.
Data Science & ML Research Intern Oct 2024 – Nov 2024
Satish Dhawan Space Centre, ISRO *Sriharikota, India*
– Reduced manual signal-review time by **~40%** by building PyTorch/TensorFlow LSTM forecasting models for **4 PSLV/SSLV telemetry streams**.
– Conducted baseline statistical profiling (stationarity, missing-value imputation) and engineered multi-source features to mitigate real-time atmospheric interference.
Intel AI4Youth Trainee, National Finalist Mar 2021 – Oct 2021
Intel Corporation (MeitY, NeGD) *Remote*
– Ranked in the **top 0.24%** of 52,000 participants by prototyping Doctor Bot, a transformer-based NLP healthcare chatbot endorsed by NITI Aayog for public deployment in Uttar Pradesh.

PROJECTS & RESEARCH

AgriSat AI: Distributed ETL & Crop-Health Analytics | Apache Spark, PySpark, scikit-learn, AWS S3, Tableau2024
– Achieved **92%+ crop-health classification accuracy** and a **35% runtime reduction** using a soft-voting ensemble (Random Forest, GBM, SVM).
– Engineered NDVI/EVI spectral features from satellite imagery and built a PySpark ETL pipeline on AWS S3, visualized via Tableau dashboards.
BDH Ablations: Bio-Inspired Dual-Circuit Language Model | PyTorch, HuggingFace, WikiText-2 2026
– Identified multiplicative gating as the most critical component (**+0.059 nats** loss on removal) via a **5-variant** ablation study of the BDH dual-circuit SSM; achieved **-0.052 nats** perplexity gain with **4x** latent compression. Paper in preparation.
Project AETHER: Lunar PSR Imaging & AI Pipeline | PyTorch, GDAL, Rasterio, GANs, HuggingFace AccelerateOngoing
– Enhanced SNR in lunar Permanently Shadowed Regions by **+86 dB** via fine-tuning a custom GAN on a self-supervised MAE pre-training stage.
– Curated and georeferenced **38,000+** training patches from **4 Chandrayaan-2 instruments** using GDAL-based geospatial ETL.
Nexus Scout: Multi-Agent Talent Acquisition Pipeline | LangGraph, LangChain, Groq API, Pydantic, Streamlit2025
– Automated personalised outreach at scale with a deterministic multi-agent LangGraph pipeline using a weighted hybrid scoring engine (60% technical fit, 40% interest alignment), reducing manual sourcing effort significantly.

TECHNICAL SKILLS

Languages: Python (Proficient), SQL, C++, Bash **CS Fundamentals:** OOP, Data Structures & Algorithms, Distributed Computing
Data Science & Analytics: Pandas, NumPy, SciPy, EDA, Statistical Modelling, Hypothesis Testing, A/B Testing, scikit-learn, XGBoost, PCA, t-SNE, UMAP, GDAL, Rasterio, Tableau, Power BI
AI & Machine Learning: PyTorch, TensorFlow, Hugging Face Transformers, State-Space Models, LSTM, GANs, Time-Series Forecasting, Bayesian Optimisation, k-fold/Stratified CV, ROC-AUC, F1, RMSE
Agentic AI & GenAI: LangChain, LangGraph, RAG Pipelines, Multi-Agent Workflows, Prompt Engineering, RLHF, LLM Evaluation, Hugging Face
Data Engineering & MLOps: Apache Spark, PySpark, Spark SQL, ETL Pipelines, Feature Engineering, MLflow, FastAPI, Docker, Kubernetes, AWS (EC2, S3), Git, Matplotlib, Seaborn

AWARDS & LEADERSHIP

Certifications: ML Engineering for Production (Andrew Ng, DeepLearning.AI) Deep Neural Networks with PyTorch (IBM/Coursera, Honours) Supervised ML: Regression & Classification (DeepLearning.AI) Deep Learning with Keras (IBM/Coursera)
Awards: DBHPS Rashtra Bhasha Praveena, All-India Champions at National Aerolympics 2020 Intel AI4Youth National Finalist (Top 125 of 52,000, MeitY & NeGD) Smart India Hackathon Finalist 2024: Social Media Fraud Prevention
Leadership: Technical Team Head, Data Science Club VIT Active Member, ML Team & AI Club VIT